

Graph Attention-Based MARL for AoI and Reliability-Aware Resource Allocation in RIS-Assisted V2X Networks



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This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFB2907700; the National Natural Science Foundation of China under Grant 62271192 and 62172142; the Major Science and Technology Projects of Longmen Laboratory under Grant 231100220300 and 231100220400; the Key Research and Development Special projects of Henan Province under Grant 241111210800; the Joint Fund Key Project of Science and Technology R&D Plan of Henan Province under Grant 225200810033, 225200810029, and 235200810040; the Key Technological Breakthrough Projects for Major Industries in Henan Province under Grant 251000220100 and 251000220300; the Scientific and Technological Key Project of Henan Province under Grant 232102210151; the Science and Technology R&D Plan Joint Fund Project of Henan Provincial under Grant 232103810040; the Central Plains Leading Talent in Scientific and Technological Innovation Program under Grant 244200510048 and 234200510018; the Strategic Research and Consulting Project of the Chinese Academy of Engineering under Grant 2023-DZ-05; the Research Project on the Mid-to-Long-Term Strategy for Industrial Technology Innovation and Development of Henan under Grant 242400411260; the Key Laboratory of Industrial Internet of Things & Networked Control, Ministry of Education under Grant 2023FF01; the Program for Science & Technology Innovation Talents in Universities of Henan Province under Grant 22HASTIT020; the Innovation and Entrepreneurship Team Project of Henan Academy of Sciences under Grant 20230201; the Henan Center for Outstanding Overseas Scientists under Grant GZS2022015; the Henan Province High-Level Foreign Experts Introduction Plan under Grant HNGD2023012; the Henan Provincial University Science and Technology Innovation Teams (Program) under Grant 25IRTSTHNO17; and the Beijing Engineering Research Center of Aerial Intelligent Remote Sensing Equipments Fund under Grant AIRSE202404; the Liaoning Natural Science Funds under Grant 2025-BS-0212; and the Dalian Science and Technology Innovation Fund under Grant 2024JB11GX001. Corresponding authors: Baofeng Ji, Boyuan Li, and Zijian Li.

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Abstract—The convergence of 6G and vehicular-to-everything (V2X) networks is pivotal for future intelligent transportation systems, yet it introduces critical challenges in resource allocation. Specifically, optimizing for the joint age of information (AoI) and reliability in reconfigurable intelligent surface (RIS)-assisted V2X networks requires navigating a fundamental trade-off. A fundamental dilemma hinders current approaches: centralized solutions offer high performance but are unscalable, while conventional decentralized multi-agent reinforcement learning (MARL) struggles with poor cooperation due to agents' partial observability. To resolve this, we propose a framework that integrates a graph attention network (GAT) into a MATD3-based centralized training and decentralized execution (CTDE) architecture. The core of this framework is an adaptive cooperative perception module, enabling each agent to weigh the importance of its neighbors and mitigate the limitations of purely local observation. Extensive simulations in a high-fidelity, 3GPP-compliant environment show that the performance gain of the proposed adaptive cooperation increases in denser network scenarios. In a challenging 40-vehicle scenario, our approach achieves a 14.8% improvement in AoI over a standard MATD3 baseline and reduces the V2V task failure rate by nearly 60%.

Index Terms—Vehicular-to-everything (V2X), age of information (AoI), reconfigurable intelligent surface (RIS), multi-agent reinforcement learning (MARL), graph attention network (GAT).

I. INTRODUCTION

THE convergence of sixth-generation (6G) wireless systems and intelligent transportation systems (ITS) is expected to play a key role in enabling connected and autonomous mobility [1]–[3]. At the heart of this transformation lies the Vehicular-to-Everything (V2X) communication network, which serves as the backbone for mission-critical applications [4]–[6]. Successfully operating this network for such applications, however, necessitates meeting a set of stringent performance requirements that present a fundamental trade-off. On one hand, Vehicle-to-Vehicle (V2V) links demand ultra-high reliability for the exchange of safety-critical messages. On the other hand, Vehicle-to-Infrastructure (V2I) uplinks, which provide real-time situational awareness to network entities, must maintain extreme timeliness, a property rigorously quantified by the Age of Information (AoI) metric [7]. Meeting these stringent requirements in the harsh and dynamic vehicular environment remains a critical and challenging problem. To this end, Reconfigurable Intelligent Surfaces (RIS) have emerged as one of the key enabling technologies for 6G, offering a promising solution [8]–[10].

By reconfiguring the wireless propagation environment, RIS can create highly reliable communication links, mitigating signal blockage and fading to enhance V2V message delivery. Concurrently, by creating virtual line-of-sight paths, RIS can significantly improve the signal-to-noise ratio for V2I uplinks, enabling more timely and reliable status updates, thereby reducing the AoI. Despite its promise, the integration of RIS into a dynamic, multi-vehicle network introduces a new layer of complexity. The continuous reconfiguration of the RIS, combined with high-speed vehicle mobility, results in a highly non-stationary and challenging propagation environment. Therefore, designing a resource allocation policy that can effectively adapt to these rapid environmental changes to jointly optimize AoI and V2V reliability remains a critical and challenging research problem.

To address the complex resource allocation challenges inherent in such dynamic wireless systems, Deep Reinforcement Learning (DRL) has emerged as a powerful, model-free paradigm [11]–[13]. Within this domain, current research efforts can be broadly categorized into two distinct paradigms: centralized single-agent DRL and distributed Multi-Agent Reinforcement Learning (MARL), each presenting a fundamental trade-off between performance guarantees and practical deployability. The centralized paradigm formulates the multi-vehicle resource allocation task as a single, large-scale Markov Decision Process, solved by a single agent with global network observability [14], [15]. While this approach appears theoretically optimal due to its global observability, but its practical viability is limited. It not only suffers from prohibitive scalability and signaling challenges, but its monolithic learning process also struggles to navigate the extremely large state-action space created by brute-force concatenation of all agents' information [16]. The monolithic learning process is prone to learning inefficiency and convergence to suboptimal policies, a challenge also validated in our experiments. These scalability and signaling challenges render it unsuitable for the dynamic, large-scale nature of real-world V2X deployments.

In stark contrast, the distributed MARL paradigm aligns with the decentralized architecture of V2X systems, where each vehicle acts as an autonomous agent [17]–[19]. Frameworks such as MATD3, often leveraging the Centralized Training and Decentralized Execution (CTDE) paradigm to mitigate the non-stationarity issue inherent in multi-agent learning [20]–[22]. However, a critical performance bottleneck persists: these agents operate on purely local observations [23]. This partial observability fundamentally limits their ability to coordinate effectively, often leading to suboptimal [24], short-sighted behaviors that fail to optimize globally coupled and temporally extended objectives like AoI. This reveals a remaining challenge: a mechanism that provides these distributed agents with the cooperative awareness necessary to transcend their local views, without sacrificing the inherent advantages of decentralized execution.

To fulfill this need, we propose a MARL framework that integrates a Graph Attention Network (GAT) into the MATD3 architecture. This approach is tailored for the intricate demands of RIS-assisted, AoI-aware V2X networks. The core of our approach is an adaptive cooperative perception module, built

upon a Graph Attention Network. Unlike conventional GNNs that treat all neighbors isotropically, our GAT-based module empowers each agent to adaptively learn the heterogeneous importance of its neighbors by attending to the most relevant entities in the real-time network state. This mechanism allows agents to extract a low-dimensional, topology-aware feature representation, effectively addressing the limitations of partial observability. In essence, we leverage centralized training and advanced cooperative mechanisms to ensure effective policy learning, while preserving the low latency, high robustness, and privacy advantages of decentralized execution.

The main contributions of this paper are summarized as follows:

- We propose a distributed MARL framework for joint AoI and reliability optimization in RIS-assisted V2X networks, leveraging the CTDE paradigm to ensure both learning stability and practical deployability. This presents a scalable and robust alternative to prevailing centralized control schemes, offering inherent resilience to physical layer impairments via its model-free design.
- While GAT has been successfully applied in various domains, its use in the highly dynamic environment of RIS-assisted V2X networks remains challenging due to the non-stationary system dynamics. To our knowledge, this is the first attempt to integrate GAT-based perception with a continuous-control actor-critic algorithm to handle the complex, hybrid action spaces inherent in this challenging scenario. We also provide a theoretical analysis establishing the convergence of the proposed algorithm and its linear execution complexity $\mathcal{O}(N)$.
- We conduct a comprehensive performance validation in a high-fidelity V2X environment based on the 3GPP TR 38.901 standard. Our evaluations show consistent improvements of the proposed framework across four key dimensions: learning efficiency, final policy performance, robustness against varying traffic loads, and scalability with network density. Notably, the performance gain of our adaptive cooperation mechanism increases in more challenging, large-scale scenarios. For instance, in a dense network of 40 vehicles, our approach reduces the average AoI by 14.8% and lowers the V2V task failure rate by nearly 60% compared to a standard MARL baseline without cooperation.

The remainder of this paper is organized as follows. Section II reviews the related work on resource allocation in V2X networks and GNN-based MARL. Section III details the system model, including the network architecture, channel models, and the formulation of the joint AoI and reliability optimization problem. In Section IV, we present our proposed GAT-MATD3 framework, elaborating on the adaptive cooperative perception module and the overall MARL algorithm design. Section V provides a comprehensive performance evaluation through extensive simulations and discusses the key findings. Finally, Section VI concludes the paper.

II. RELATED WORK

This section reviews the literature from three key perspectives that form the foundation of our work: resource allocation

in RIS-assisted vehicular networks, the evolution of MARL for wireless communications, and the pivotal role of Graph Neural Networks (GNNs) in enabling cooperative MARL.

The integration of RIS into V2X networks has recently garnered significant attention as a promising technique to enhance wireless link quality and combat the adverse effects of the highly dynamic propagation environment [25], [26]. Early research in this domain primarily focused on applying traditional, model-based optimization techniques to the resource allocation problem. These works typically formulate the joint design of active beamforming at the base station and passive beamforming at the RIS as a non-convex optimization problem, which is then solved using iterative algorithms such as alternating optimization (AO), block coordinate descent (BCD), or successive convex approximation (SCA) [27], [28]. Recently, the scope of RIS research has expanded beyond terrestrial surfaces. For instance, STAR-RIS has also been explored in drone-assisted 5G frameworks to enable flexible 3D resource scheduling and full-space coverage [29], primarily under model-based or centralized optimization assumptions. However, the efficacy of these model-based approaches is fundamentally predicated on the availability of accurate and, often, instantaneous channel state information (CSI). This assumption is particularly problematic in V2X scenarios, where high vehicle mobility and complex urban multipath environments make frequent and precise CSI acquisition practically infeasible [30]. The performance of these methods degrades sharply in the presence of imperfect or outdated CSI, and their iterative nature can lead to high computational complexity, rendering them unsuitable for the low-latency decision-making required in V2X applications. To address these limitations, a growing body of work has begun to explore DRL as a powerful, model-free alternative. DRL agents can learn effective resource allocation policies directly from interaction with the environment, bypassing the need for an explicit channel model or perfect CSI. Notably, the work by Qi et al. [14] pioneered an AoI-aware resource allocation scheme using a centralized Soft Actor-Critic (SAC) agent, demonstrating the potential of DRL to handle the joint optimization of complex metrics like AoI in RIS-assisted IoV networks. While seminal, this work, along with similar centralized DRL approaches [31], still faces the fundamental scalability and practical deployability challenges inherent to centralized control, as discussed in our introduction. This leaves the critical problem of designing a scalable MARL framework with explicit cooperation mechanisms largely open.

The inherent distributed nature of V2X networks makes MARL a natural and scalable framework for decentralized resource allocation. Early MARL approaches, which often treated agents as independent learners (e.g., Independent Q-Learning), struggled with the severe non-stationarity of the learning environment, where the concurrent policy updates of agents render the environment unstable from any single agent's perspective [24]. This fundamental challenge spurred the development of the CTDE paradigm, which has since become the de facto standard in modern MARL research [32]. Within the CTDE framework, off-policy actor-critic algorithms have proven particularly effective. The Multi-Agent Deep

Deterministic Policy Gradient (MADDPG) algorithm [33] was a seminal work in this area, introducing a centralized critic that utilizes the joint observations and actions of all agents to produce a stable training signal. Building upon this, the Multi-Agent Twin Delayed DDPG (MATD3) algorithm [34] was proposed to mitigate the Q-value overestimation bias inherent in MADDPG, further enhancing learning stability and performance. Owing to their robustness and effectiveness, CTDE-based actor-critic methods, particularly MATD3, have been widely and successfully applied to a variety of complex wireless resource allocation problems, such as in RAN slicing and UAV networks [20], [22]. Concurrent with CTDE, distributed paradigms exploring privacy and efficiency have also emerged. Notably, Quantized Federated Reinforcement Learning (Q-FRL) has been proposed to optimize energy-latency tradeoffs in carbon-intelligent systems [35], while Quantum Federated RL has been leveraged to handle complex joint mode selection under different system assumptions in STAR-RIS-aided networks [36], with a primary emphasis on federated or quantum optimization rather than cooperative perception. Despite its success in ensuring learning stability, the CTDE paradigm does not, in itself, solve the fundamental problem of partial observability at the agent level. During the crucial decentralized execution phase, each agent's actor must still make decisions based on its own local, and often insufficient, observation. This "information island" problem remains a critical performance bottleneck. Without an explicit mechanism to reason about the real-time states and intentions of their neighbors, agents are fundamentally limited to learning reactive policies, preventing them from forming the sophisticated, proactive cooperative strategies necessary for optimizing complex, system-level objectives like AoI.

To overcome the fundamental challenge of partial observability in MARL, a promising line of research has focused on leveraging GNNs as an explicit communication and cooperative perception mechanism [39]. By modeling the multi-agent system as a graph, in which agents are nodes and their relationships are edges, GNNs enable agents to reason about the network topology and aggregate information from their neighbors, thereby forming a richer, more comprehensive state representation for decision-making. Initial applications of this paradigm in vehicular networks utilized foundational GNN architectures such as GraphSAGE [37], demonstrating the general viability of using GNNs to enhance cooperation in V2X resource allocation. However, the isotropic aggregation mechanisms of these early GNNs (e.g., mean-pooling in GraphSAGE) treat all neighbors with equal importance. However, this assumption is overly simplistic in V2X environments. In practice, inter-vehicle influence is highly heterogeneous and dynamic. Recognizing this limitation, a more advanced line of research has centered on GAT [40]. GAT introduces a self-attention mechanism that allows each agent to adaptively weigh the importance of its neighbors, enabling a more intelligent and selective information aggregation. This attention-based approach has proven highly effective in other complex wireless MARL scenarios, such as in network slicing where agents learn to attend to the most informative neighboring base stations [38]. While the progression to attention-based GNNs

TABLE I: Comparison of the Proposed Work With Related Work

Reference	RIS	V2X	Architectural Paradigms			Objective Function				Cooperation Mechanism	DRL-based Approach
			Centralized	Decentralized	CTDE	AoI	Reliability	Sum Rate	Energy Eff.	Joint Opt.	
[14]	✓	✓	✓			✓	✓			✓	✓
[27]	✓	✓	✓				✓	✓		✓	
[28]	✓		✓				✓				
[31]	✓		✓				✓				✓
[15]		✓		✓			✓	✓		✓	✓
[17]		✓		✓			✓		✓	✓	✓
[19]		✓		✓			✓	✓		✓	✓
[20]					✓		✓	✓		✓	✓
[22]	✓	✓			✓				✓		✓
[34]					✓						✓
[37]		✓			✓		✓	✓		GraphSAGE	✓
[38]					✓		✓		✓	GAT	✓
Proposed Work	✓	✓			✓	✓	✓			GATv2	✓

marks a significant methodological advance, their application in MARL has largely been confined to domains with either simplified physical models or discrete action spaces. The critical challenge of how to leverage such advanced cooperative perception mechanisms within actor-critic frameworks to master the complex, hybrid action spaces inherent in high-stakes cyber-physical systems (such as our RIS-assisted V2X network) remains largely unaddressed. Simply integrating a GNN is insufficient; a holistic co-design of the perception module and the decision-making architecture is required. Our work addresses this fundamental gap by proposing a new paradigm for GNN-enhanced MARL. We introduce a bespoke adaptive cooperative perception module, built upon a GAT, and demonstrate how to deeply integrate it with a continuous-control MARL algorithm (MATD3). Our approach integrates the GAT-based perception pipeline with actor-critic learning dynamics in a co-design manner. To visually summarize our contributions and position our work within the existing literature, a detailed comparison of key features is presented in Table I.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we present the system model for our RIS-assisted multi-agent V2X network. We first describe the network scenario, followed by the vehicular mobility model. Then, we detail the channel models for both V2V and V2I links, considering the presence of the RIS and realistic propagation effects. Finally, we define the key performance metrics and formulate the overall resource allocation problem as a stochastic optimization task.

A. Scenario

As illustrated in Fig. 1, we consider a V2X network deployed in a canonical urban canyon environment, which is representative of dense city blocks where wireless propagation is challenging. The system comprises a set of N mobile vehicles, denoted by $\mathcal{N} = \{1, 2, \dots, N\}$, a single Base Station (BS), and a single RIS. In our Multi-Agent Reinforcement Learning (MARL) framework, each vehicle $i \in \mathcal{N}$ is modeled as an autonomous, decision-making agent equipped with a single-antenna transceiver at a height of H_{veh} . The BS is strategically placed at a fixed 3D coordinate $\mathbf{p}_B$ with an antenna height of

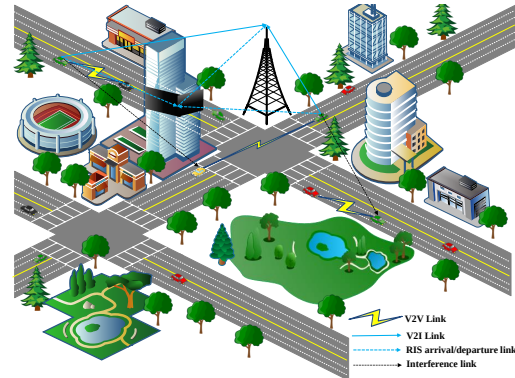


Fig. 1: System architecture of the RIS-assisted V2X network.

H_{BS} . The passive RIS, composed of M reflecting elements, is situated on a building facade at $\mathbf{p}_R$ with a height of H_{RIS} . The network is designed to support two concurrent uplink communication services over a spectrum centered at f_c . The spectrum is divided into $K = N$ orthogonal Resource Blocks (RBs), indexed by $\mathcal{K} = \{1, 2, \dots, K\}$, each with a bandwidth of W . The first service is the V2I link, where each vehicle i periodically transmits its status updates to the BS, for which the primary performance objective is to maintain timeliness, as quantified by the AoI. The second service is the V2V link, where each vehicle i establishes a connection with its single ($N_{\text{neighbor}} = 1$) nearest neighbor j , for which the core performance requirement is high reliability.

B. Mobility Model

We model the vehicular mobility within a discrete-time framework, where the system state evolves in time steps of duration Δt . The simulation environment represents a large urban intersection formed by the crossing of two multi-lane, bidirectional roads. This Manhattan-style grid topology is widely adopted in V2X network simulations for its ability to capture the structured and constrained nature of urban traffic flow [41].

At the beginning of each simulation episode, each vehicle $i \in \mathcal{N}$ is initialized with a random 2D position $\mathbf{p}_i(0) = [x_i(0), y_i(0)]$ on one of the lanes, a random cardinal direction, and a constant speed v_i drawn uniformly from a predefined

range $[v_{\min}, v_{\max}]$. The vehicle's movement is governed by a set of rules that emulate urban driving behavior. For a vehicle traveling straight, its position at time slot $n + 1$ is updated based on its constant velocity vector $\mathbf{v}_i$:

$$\mathbf{p}_i(n+1) = \mathbf{p}_i(n) + \mathbf{v}_i \cdot \Delta t. \quad (1)$$

To capture more realistic traffic dynamics, we model the vehicle's behavior upon approaching the intersection area. When a vehicle's trajectory is about to cross into the defined intersection zone, a stochastic decision is made. With a predefined probability P_{turn} , the vehicle will execute a turn; otherwise, it will proceed straight according to (1). If a turn is executed, the vehicle is randomly assigned to a new lane in the perpendicular direction. For instance, for a vehicle i traveling upward and deciding to turn left, its new position is calculated as:

$$x_i(n+1) = x_i(n) - (v_i \cdot \Delta t - \Delta y), \quad (2)$$

$$y_i(n+1) = y_{\text{lane}}^{\text{new}}, \quad (3)$$

where $y_{\text{lane}}^{\text{new}}$ is the coordinate of the newly selected horizontal lane and Δy is the remaining distance to the intersection boundary. A similar model is applied for other turning directions. The boundaries of the simulation area are modeled as toroidal to maintain a constant vehicle density and avoid edge effects, ensuring statistical stationarity of the traffic flow over long-term simulations. While this model abstracts away microscopic behaviors such as acceleration and lane changes for tractability, it effectively captures the first-order mobility dynamics and topological changes that are the primary drivers of the time-varying channel conditions.

C. Communication Model

In our 3D Cartesian coordinate system, the channel between any two nodes incorporates both large-scale and small-scale fading effects to ensure a high-fidelity representation of urban V2X propagation. The positions of the Base Station (BS), Reconfigurable Intelligent Surface (RIS), and vehicle i at time slot n are denoted by $\mathbf{p}_B$, $\mathbf{p}_R$, and $\mathbf{p}_i^n$, respectively. For large-scale effects, we adopt the 3GPP TR 38.901 standard for the UMi-Street Canyon scenario [42], which combines a linear path loss gain $\rho(d)$ and a shadow fading gain S^n . The path loss model distinguishes between Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) conditions based on a probabilistic function of the 2D distance. Shadow fading is modeled as a spatially correlated log-normal process. Small-scale fading is modeled as Rayleigh fading, represented by a circularly-symmetric complex Gaussian random variable $g^n \sim \mathcal{CN}(0, 1)$.

The V2V link consists solely of the direct path between the transmitting vehicle i and the receiving vehicle j . Its instantaneous complex channel gain at time slot n , $h_{i,j}^n$, is expressed as:

$$h_{i,j}^n = \sqrt{\rho(d_{i,j}^n) \cdot S_{i,j}^n} \cdot g_{i,j}^n, \quad (4)$$

where $\rho(d_{i,j}^n)$ and $S_{i,j}^n$ are the gains from path loss and shadow fading, respectively.

The uplink channel from vehicle i to the BS is a coherent superposition of the direct path, $h_{i,B,d}^n$, and the RIS-reflected

path, $h_{i,B,r}^n$. Thus, its total equivalent complex channel gain, $h_{i,B}^n$, is given by:

$$h_{i,B}^n = h_{i,B,d}^n + h_{i,B,r}^n. \quad (5)$$

The direct path gain is modeled similarly to the V2V channel in (4), but using V2I-specific parameters. The reflected path gain represents a cascaded channel and is formulated as:

$$h_{i,B,r}^n = (\mathbf{h}_{R,B}^n)^H \Theta^n \mathbf{h}_i^n. \quad (6)$$

Here, $\mathbf{h}_i^n \in \mathbb{C}^{M \times 1}$ and $\mathbf{h}_{R,B}^n \in \mathbb{C}^{M \times 1}$ are the channel vectors for the vehicle-RIS and RIS-BS links, respectively. These vectors incorporate both large-scale fading gains and the array steering vectors, which are functions of the Angles of Arrival (AoA) and Departure (AoD). The term $\Theta^n = \text{diag}(e^{j\theta_1^n}, \dots, e^{j\theta_M^n})$ is the diagonal phase-shift matrix of the RIS.

Based on the channel models above, we formulate the SINR for both links. Let P_i^n be the transmit power and k_i^n be the RB index chosen by vehicle i for its V2V communication. For a generic V2I user u transmitting on RB l with a fixed uplink power P_u^{V2I} , the received SINR at the BS, $\gamma_{u,B}^{I,n}[l]$, is given by:

$$\gamma_{u,B}^{I,n}[l] = \frac{P_u^{\text{V2I}} |h_{u,B}^n|^2}{\sum_{j \in \mathcal{N}, k_j^n = l} P_j^n |h_{j,B}^n|^2 + \sigma^2}, \quad (7)$$

where the summation in the denominator represents the co-channel interference from all V2V users j reusing RB l , and σ^2 is the AWGN power. For the V2V pair (i, j) , where transmitter i uses RB $l = k_i^n$, the SINR at receiver j , $\gamma_{i,j}^{V,n}[l]$, is:

$$\gamma_{i,j}^{V,n}[l] = \frac{P_i^n |h_{i,j}^n|^2}{I_{\text{BS}}^n[l] + I_{\text{V2V}}^n[l] + \sigma^2}. \quad (8)$$

The total interference in the denominator consists of two components. The term $I_{\text{BS}}^n[l]$ represents interference from the BS. We assume channel reciprocity, which is a valid assumption in the Time Division Duplex (TDD) systems commonly envisioned for V2X communications. Consequently, the downlink channel from the BS to the V2V receiver j can be approximated by the uplink channel $h_{j,B}^n$. Thus, the interference is modeled as $I_{\text{BS}}^n[l] = P_{\text{BS}} |h_{j,B}^n|^2$, where P_{BS} is the transmit power of the BS on that RB. The term $I_{\text{V2V}}^n[l] = \sum_{j' \in \mathcal{N}, j' \neq i, k_{j'}^n = l} P_{j'}^n |h_{j',j}^n|^2$ is the aggregate interference from other co-channel V2V transmitters.

Remark 1. The block-fading assumption aligns with the MDP decision interval. Regarding hardware impairments, they can be abstracted as state observation noise in our model-free DRL framework. As shown in Sec. V, the agent inherently learns to adapt to these uncertainties, ensuring the scheme's validity without explicit hardware modeling.

D. AoI and Reliability Model

In vehicular networks, the timeliness of V2I communication is vital for maintaining precise situational awareness, while the reliability of V2V communication is directly linked to driving safety. To precisely characterize these distinct Quality-of-Service (QoS) requirements, we introduce the Age of Information (AoI) to measure the data freshness of V2I links and

formulate a payload transmission success probability model to assess the reliability of V2V links.

The performance of the V2I uplink is quantified by the AoI, which measures the freshness of the status updates received at the BS. Let A_u^n denote the AoI of V2I user u at the beginning of time slot n . A successful transmission is deemed to occur if the achievable data rate, $R_u^n[l] = W \log_2(1 + \gamma_{u,B}^{I,n}[l])$, exceeds a predefined minimum threshold R_{th}^I . The AoI evolves over time according to the following rule:

$$A_u^{n+1} = \begin{cases} \Delta t, & \text{if } R_u^n[l] \geq R_{th}^I \\ \min(A_u^n + \Delta t, A_{\max}), & \text{otherwise} \end{cases} \quad (9)$$

where Δt is the duration of a time slot, and $A_{\max}$ is the maximum permissible AoI.

For the V2V links, the key performance metric is reliability, which we model as the payload transmission success probability. We consider a scenario where each V2V pair (i, j) is tasked with transmitting a data payload of size D_V bits within a maximum tolerable delay of T_V . The transmission success probability for the pair $\mathcal{P}_{\text{succ}}(i, j)$ is formulated as:

$$\mathcal{P}_{\text{succ}}(i, j) = \Pr \left(\sum_{n=1}^{T_V/\Delta t} W \log_2(1 + \gamma_{i,j}^{V,n}[k_i^n]) \cdot \Delta t \geq D_V \right). \quad (10)$$

This metric captures the link's ability to satisfy the data demand under the joint effects of channel dynamics, resource allocation decisions, and interference.

While (10) provides a formal definition of reliability, for the purpose of formulating a tractable reinforcement learning problem, we utilize an instantaneous proxy metric in our objective function. Specifically, we use the proportion of remaining V2V data payload across all links as an indicator of reliability failure. Therefore, the term $(1 - \mathcal{P}_{\text{succ}}^n)$ in our objective function is directly represented by the normalized sum of remaining V2V demand at time slot n . Minimizing this value encourages the agent to complete V2V transmissions as quickly as possible, thereby maximizing reliability.

E. Problem Formulation

Grounded in the system model, our objective is to devise an intelligent resource allocation policy, denoted by π . A key aspect of our system modeling is the management of the RIS phase-shift matrix, Θ^n . In this work, to decouple the problem and maintain a clear focus on the intricate challenge of inter-agent cooperation, we model the RIS phase-shift control as part of the environmental dynamics. Specifically, the RIS is assumed to be managed by a low-complexity heuristic controller that dynamically steers its beam towards the centroid of the vehicle cluster. This modeling choice allows us to establish a stable yet adaptive environmental, against which the learning and cooperative capabilities of the decentralized vehicular agents can be rigorously and unambiguously evaluated.

The policy π is therefore dedicated to jointly determining the optimal transmit power P_i^n and spectrum resource block k_i^n for each V2V link. We address the dual objective of minimizing V2I AoI while maximizing V2V reliability by

formulating it as the minimization of a single, weighted-sum cost function. This leads to the following stochastic optimization problem (P1):

$$\min_{\pi} \mathbb{E} \left[\sum_{n=0}^{N_{\text{total}}} \gamma^n \left(w_A \sum_{u \in \mathcal{N}} A_u^n + w_P (1 - \mathcal{P}_{\text{succ}}^n) \right) \right] \quad (11)$$

subject to:

$$0 \leq P_i^n \leq P_{\max}, \quad \forall i \in \mathcal{N}, \forall n \quad (C1)$$

$$\mathbb{E}[P_i^n] \leq P_{\text{avg}}, \quad \forall i \in \mathcal{N} \quad (C2)$$

$$\sum_{l \in \mathcal{K}} \mathbb{I}(k_i^n = l) = 1, \quad \forall i \in \mathcal{N}, \forall n \quad (C3)$$

$$\sum_{i \in \mathcal{N}} \mathbb{I}(k_i^n = l) \leq K_{\text{reuse}}, \quad \forall l \in \mathcal{K}, \forall n \quad (C4)$$

$$j = \arg \min_{j' \in \mathcal{N}, j' \neq i} \|\mathbf{p}_i^n - \mathbf{p}_{j'}^n\|, \quad \forall (i, j) \in \mathcal{P}_V^n \quad (C5)$$

The objective function in (11) seeks to minimize the expected cumulative weighted cost over an episode of N_{total} time slots. The expectation $\mathbb{E}[\cdot]$ is taken over the system stochasticity and the policy π . The term $\gamma \in [0, 1]$ is the discount factor, while w_A and w_P are non-negative weights that balance the trade-off between the AoI of V2I users, A_u^n , and the V2V success captured by the proportion of successful links, $\mathcal{P}_{\text{succ}}^n$.

The policy π must adhere to several operational constraints. Constraint (C1) enforces the peak power limit $P_{\max}$. Energy efficiency is ensured by (C2), which imposes a long-term average power budget P_{avg} . The unique channel assignment for each transmitter is guaranteed by (C3), where $\mathbb{I}(\cdot)$ is the indicator function. To manage co-channel interference, (C4) restricts the spectrum reuse to a maximum of K_{reuse} links per RB. Finally, (C5) specifies the nearest-neighbor pairing protocol, where $\mathcal{P}_V^n$ is the set of active V2V pairs at time slot n . The formulated problem (P1) is a non-convex stochastic program with a high-dimensional, mixed-integer action space and complex, non-stationary dynamics, rendering traditional optimization methods impractical. The intractability is compounded by the time-varying topology in (C5) and the long-term constraint in (C2). This motivates our adoption of a deep reinforcement learning (RL) framework, for which we reformulate (P1) as a Markov Decision Process (MDP) in the next section.

Remark 2. The Nearest-Neighbor pairing in (C5) and the fixed spectrum reuse limit in (C4) are adopted to capture the stringent requirements of safety-critical V2X scenarios (e.g., collision avoidance) and physical layer interference limits, respectively. While these settings provide a controlled environment to rigorously evaluate the cooperative perception mechanism, the proposed GAT-MATD3 framework is inherently adaptable to alternative policies. Since the GAT module processes the communication graph based on its topological structure rather than fixed identifiers, adapting to different pairing rules (e.g., QoS-based pairing) only requires updating the input adjacency matrix. Similarly, dynamic spectrum reuse can be achieved by reformulating the hard constraint (C4) into a soft penalty within the reward function.

IV. METHODOLOGY

To address the complex and dynamic resource allocation optimization problem established in Chapter IV, this chapter details our proposed solution based on MARL. We begin by rigorously formulating the problem as a Multi-Agent Markov Decision Process (MMDP). Subsequently, we will delve into the specific architecture and training mechanisms of our proposed GAT-MATD3 algorithm. Finally, an analysis of the algorithm's computational complexity will be presented.

A. Problem Formulation as a Multi-Agent MDP

To cast the aforementioned optimization problem (P1) into a framework suitable for reinforcement learning solutions, we rigorously model it as a decentralized partially observable Markov decision process (Dec-POMDP). This framework naturally captures the distributed nature of V2X networks, where each vehicle acts as an independent decision-making unit. For our problem, the core elements of the Dec-POMDP are the tuple $\langle \mathcal{N}, \{\Omega_i\}, \{\mathcal{A}_i\}, \{\mathcal{R}_i\}, \gamma \rangle$, defined as follows:

1) Agents: The set of agents consists of the N vehicles in the scenario, denoted by $\mathcal{N}$, where each vehicle $i \in \mathcal{N}$ is an autonomous decision-making agent.

2) Observation Space: At each timestep n , each agent i receives an observation $o_i^n \in \Omega_i$ that reflects its local environment. This observation vector is constructed by concatenating four key components and can be formally expressed as:

$$o_i^n = [\mathbf{g}_{i,B}^n, \mathbf{g}_{i,j}^n, A_i^n, d_{i,\text{rem}}^n], \quad (12)$$

where $\mathbf{g}_{i,B}^n \in \mathbb{R}^K$ is the V2I uplink channel gain vector, locally acquired via TDD channel reciprocity using downlink pilots; $\mathbf{g}_{i,j}^n \in \mathbb{R}^K$ is the V2V channel gain vector to the nearest neighbor j , obtained via C-V2X Mode 4 sidelink sensing (e.g., DMRS measurements); $A_i^n \in \mathbb{R}^+$ is the current Age of Information for agent i 's V2I link; and $d_{i,\text{rem}}^n$ is the normalized remaining data demand ratio for agent i 's current V2V link, defined as $d_{i,\text{rem}}^n = D_{i,\text{rem}}^n / D_V$, where $D_{i,\text{rem}}^n$ is the remaining data in bits and D_V is the initial payload size. These components rely on local measurements, which supports the decentralized execution of the framework. All these observation components are further normalized before being input to the neural networks.

3) Action Space: The action $a_i^n \in \mathcal{A}_i$ for each agent i is a two-dimensional continuous vector, $a_i^n = [a_{i,P}^n, a_{i,k}^n]$, whose elements are normalized to the range $[-1, 1]$ by the actor network's output layer. This abstract action vector is subsequently decoded into two concrete resource allocation decisions through deterministic transformations. The first element, $a_{i,P}^n$, determines the V2V transmit power P_i^n via a linear mapping:

$$P_i^n = \frac{a_{i,P}^n + 1}{2}(P_{\max} - P_{\min}) + P_{\min}, \quad (13)$$

where the resulting power P_i^n is interpreted in dBm within the practical range of $[P_{\min}, P_{\max}]$. The second element, $a_{i,k}^n$, is discretized to select a spectrum resource block k_i^n from the

K available options. This is achieved by first normalizing the action to the range $[0, 1)$ and then applying the floor function:

$$k_i^n = \left\lfloor \frac{a_{i,k}^n + 1}{2} \cdot K \right\rfloor + 1, \quad (14)$$

where $\lfloor \cdot \rfloor$ denotes the floor function. Note that for implementation, a small constant ϵ is typically subtracted from K to prevent out-of-bounds indexing when $a_{i,k}^n = 1$.

To circumvent the curse of dimensionality inherent to large-scale discrete action spaces, we employ a compact scalar-mapping strategy rather than high-dimensional categorical parameterizations. By utilizing a continuous-input Critic to preserve gradient information, this design strikes a robust balance between scalability and learning stability, as substantiated by the comparative convergence analysis in Fig. 12.

4) Reward Function: To guide the agents in learning to address the dual objectives defined in the optimization problem (P1), we meticulously design a reward function that directly reflects the instantaneous system cost. At timestep n , agent i 's reward r_i^n is the weighted sum of penalties for its normalized AoI and normalized remaining V2V demand:

$$r_i^n = - \left(w_A \cdot \frac{A_i^n}{A_{\max}} + w_P \cdot \frac{D_{i,\text{rem}}^n}{D_V} \right), \quad (15)$$

where w_A and w_P are hyperparameters that balance the two objectives. The AoI and demand terms are normalized by $A_{\max}$ and D_V to the unit interval $[0, 1]$, ensuring scale comparability and preventing gradient dominance by larger-magnitude objectives. By maximizing the long-term discounted cumulative sum of this reward, agents are incentivized to learn a policy that simultaneously reduces information age and rapidly completes V2V data transmissions. The remaining demand term serves as a dense surrogate to mitigate the sparse reward problem inherent in the binary reliability metric of Eq. (10). While not a strict theoretical equivalent, this proxy introduces an inductive bias that steers the policy to maximize instantaneous throughput, inducing policies that satisfy the deadline constraint under the considered system settings. Power consumption is managed via the constraints (C1) and (C2) in problem (P1), rather than being incorporated as a penalty term in the reward function. This design grants the agents greater flexibility to leverage power resources for optimizing the core communication metrics.

5) Discount Factor: The discount factor, $\gamma \in [0, 1)$, is introduced in the objective function of problem (P1) to balance the trade-off between immediate and future rewards. Its specific value is provided in the experimental setup section.

B. GAT-MATD3 Algorithm Architecture

The proposed GAT-MATD3 algorithm is founded upon the CTDE paradigm. This approach leverages global information during the offline training phase to facilitate stable and efficient learning, while permitting low-latency, decentralized decision-making during online execution. Our methodology represents a substantive integration of an upgraded Graph Attention Network (GATv2) within the MATD3 framework. The core innovation lies in the GAT-augmented actor network, which

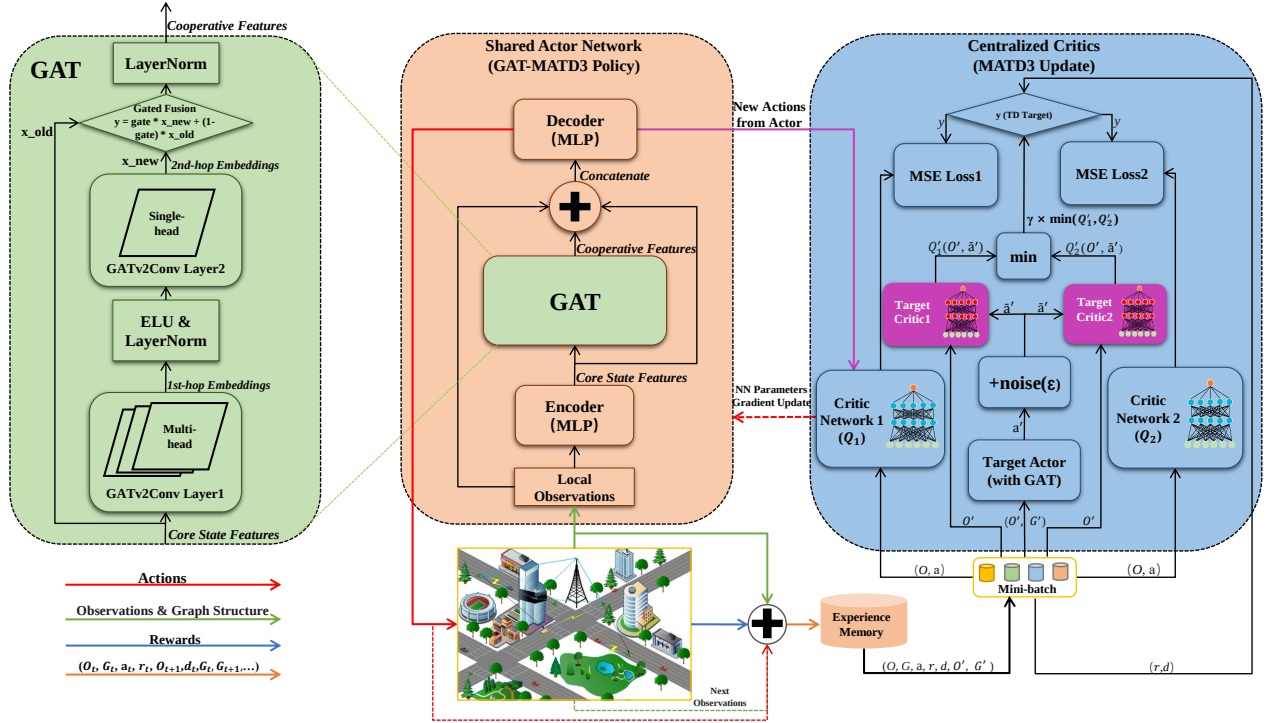


Fig. 2: Overview of the GAT-MATD3 algorithm architecture. The left panel shows the two-layer GAT structure, the center illustrates the actor network flow, and the right panel depicts the critic update process with twin critics and policy smoothing.

enables agents to perceive and adapt to the dynamic network topology. To ensure robust training, the algorithm preserves the foundational techniques of MATD3. The comprehensive architecture of this approach is illustrated in Fig. 2.

The architecture of the shared actor network, depicted in the central panel of Fig. 2, is composed of three sequential modules. First, an encoder module transforms the raw, high-dimensional local observation vector $o_i \in \mathbb{R}^{D_o}$ of each agent i into a dense, low-dimensional latent representation, termed the *Core State Feature* $h_i \in \mathbb{R}^{D_h}$. This is accomplished via a shared multi-layer perceptron (MLP) with layer normalization, as per the following transformation:

$$h_i = f_{\text{enc}}(o_i; \phi_{\text{enc}}). \quad (16)$$

This process yields a set of node embeddings $\mathbf{h} = \{h_1, \dots, h_N\}$ that serve as the input for graph-based processing. Next, to model inter-agent interactions, we dynamically construct a communication graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ at each timestep. The resulting core state features $\mathbf{h}$ are then processed by the GAT module, whose internal structure is detailed in the left panel of Fig. 2.

The GAT employs a self-attention mechanism, allowing each agent to dynamically assign importance weights to its neighbors. We utilize the more expressive **GATv2** [43] architecture, where the attention coefficient e_{ij} , signifying the importance of agent j 's features to agent i , is computed as:

$$e_{ij} = \mathbf{w}_{\text{attn}}^T \text{LeakyReLU}(\mathbf{W}_i h_i + \mathbf{W}_j h_j), \quad (17)$$

where $\mathbf{W}_i, \mathbf{W}_j$ are learnable weight matrices and $\mathbf{w}_{\text{attn}}$ is a learnable weight vector. This dynamic attention mechanism allows the importance of a neighbor to depend on the querying

agent itself. These scores are normalized using the softmax function to obtain the final attention weights α_{ij} .

As shown in the diagram, the first layer employs a multi-head attention mechanism to produce *1st-hop Embeddings*. After an ELU activation and layer normalization, a single-head second layer generates *2nd-hop Embeddings*.

A crucial element is the gated residual connection, which adaptively fuses the input core state features (h_i) with the second-layer output (h'_i). This mitigates the over-smoothing problem and results in the final *Cooperative Features* c_i :

$$\begin{aligned} g_i &= \sigma(\mathbf{W}_g[h'_i \| h_i]), \\ c_i &= \text{LayerNorm}((1 - g_i) \odot h_i + g_i \odot h'_i), \end{aligned} \quad (18)$$

where g_i is a learnable gate computed by a small neural network with weights $\mathbf{W}_g$, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication.

Finally, the action is produced by a decoder module. As shown by the 'Concatenate' block in Fig. 2, the decoder's input is a comprehensive feature vector formed by the concatenation of the original observation o_i , the core state feature h_i , and the cooperative feature c_i . The output is passed through a tanh function to produce the normalized action $a_i \in [-1, 1]^{D_a}$:

$$a_i = \tanh(f_{\text{dec}}([o_i \| h_i \| c_i]; \phi_{\text{dec}})). \quad (19)$$

The training mechanism, illustrated in the right panel of Fig. 2, is centralized. For each agent i , we maintain a pair of independent critic networks, $Q_{\theta_{i,1}}$ and $Q_{\theta_{i,2}}$. These critics take the joint observations $\mathbf{o}$ and joint actions $\mathbf{a}$ as input to estimate the action-value function $Q_i(\mathbf{o}, \mathbf{a})$. The networks are updated by sampling mini-batches of transitions from the replay buffer. The critic networks are trained by minimizing

the Mean Squared Bellman Error (MSBE). As depicted, the target value y_i is computed by first feeding the next state $\mathbf{o}'$ and graph structure $\mathcal{G}'$ into the target actor network to obtain the next action $\mathbf{a}'$. Noise is added to this action, and the result $\bar{\mathbf{a}}'$ is passed to the twin target critics. The minimum of the two target Q-values is used to form the TD target:

$$y_i = r_i + \gamma(1 - d) \min_{k=1,2} Q_{\theta'_{i,k}}(\mathbf{o}', \bar{\mathbf{a}}'). \quad (20)$$

The loss for each critic, $L(\theta_{i,k})$, is then calculated as the mean squared error between its predicted Q-value and the target value y_i . This loss is formally expressed as:

$$L(\theta_{i,k}) = \mathbb{E}_{(\mathbf{o}, \mathbf{a}, y_i) \sim \mathcal{D}} [(Q_{\theta_{i,k}}(\mathbf{o}, \mathbf{a}) - y_i)^2]. \quad (21)$$

The parameters $\theta_{i,k}$ of each critic network are then updated by descending the gradient of this loss.

The shared actor network is updated with a lower frequency, as dictated by the policy delay hyperparameter d_{policy} . To facilitate stable learning in our dynamic multi-agent setting, our framework implements an iterative procedure for the actor update. Specifically, within a single training step's update phase, we loop through all agents. For each agent $i \in \mathcal{N}$, its individual policy gradient is computed and immediately used to perform a gradient update on the shared actor parameters. The loss function driving this update for agent i is defined as:

$$J_i(\phi) = -\mathbb{E}_{\mathbf{o} \sim \mathcal{D}} [Q_{\theta_{i,1}}(\mathbf{o}, a_1, \dots, \mu_{\phi_i}(o_i, \mathcal{G}_t), \dots, a_N)], \quad (22)$$

where the actions a_j for $j \neq i$ are outputs from the actor network but are detached from the computation graph for this specific update. The shared actor parameters ϕ are then updated by one step of gradient ascent:

$$\phi \leftarrow \phi + \alpha_A \nabla_{\phi} J_i(\phi). \quad (23)$$

This process, sequentially refining the shared policy based on each agent's objective, proved to be a robust and effective strategy in our complex environment, as detailed in Algorithm 1.

Finally, to ensure stability, all target network parameters are updated slowly via Polyak averaging, where $\tau \ll 1$ is the update rate:

$$\phi' \leftarrow \tau \phi + (1 - \tau) \phi', \quad (24)$$

$$\theta'_{i,k} \leftarrow \tau \theta_{i,k} + (1 - \tau) \theta'_{i,k}, \quad \text{for } k \in \{1, 2\}. \quad (25)$$

The complete training procedure, which integrates environment interaction, experience replay, and coordinated updates of the actor and critic networks, is summarized in Algorithm 1.

C. Stability and Convergence Analysis

To formally ground our algorithm, this section provides a concise theoretical analysis of the stability and convergence of GAT-MATD3. We demonstrate that integrating the GAT module preserves and enhances the convergence guarantees of the underlying multi-agent actor-critic framework. Our analysis is built upon standard assumptions and established results from stochastic approximation theory [44], [45]. For rigor, we provide the detailed proofs in Appendix A and B.

1) *Core Assumptions*: Our analysis rests on a set of standard, formal assumptions that collectively ensure the asymptotic convergence of GAT-MATD3. These include standard conditions on learning rates (Robbins-Monro), a martingale difference noise model for sampling, and the smoothness and boundedness of the network function approximators. A key assumption is the Lipschitz continuity of our networks, which the GAT module inherits as a composition of Lipschitz continuous layers. This ensures that the cooperative perception mechanism is robust and does not produce erratic updates, vital for stable learning in our dynamic MARL setting.

2) *Convergence Results*: Building upon these foundations, we can state the main convergence results.

Theorem 1 (Stable Value Estimation): Under standard assumptions (detailed in Appendix A), each critic network's parameters θ_t converge to a bounded neighborhood of the true action-value function Q^μ . The size of this neighborhood is determined by the network's function approximation error and a bounded algorithmic bias. *Proof Sketch*: The critic update is analyzed as a stochastic approximation process. The core stability is derived from the TD3 mechanisms: the slow-updating target network provides a quasi-static learning target, while clipped double Q-learning mitigates overestimation bias. These ensure the underlying ODE system is stable. The convergence to a neighborhood rather than a point is a consequence of using non-linear function approximators and the inherent algorithmic bias.

Theorem 2 (Convergent Resource Allocation Policy): Given a stable critic from Theorem 1, and under two-timescale update conditions, the GAT-enhanced actor's resource allocation policy μ_ϕ converges to a bounded neighborhood of a stationary point of the policy performance objective. *Proof Sketch*: We leverage two-timescale stochastic approximation theory. The critic operates on a fast timescale, providing a stable value estimate for the current policy. The actor operates on a slow timescale, performing gradient ascent on the value landscape estimated by the critic. In our framework, this timescale separation is realized by the **delayed policy update** mechanism from TD3. The Lipschitz continuity of the entire GAT-enhanced actor is crucial, as it guarantees a smooth policy landscape and bounded gradients, ensuring the underlying ODE of the actor's update converges. The final convergence is to a neighborhood whose size is proportional to the critic's estimation error.

Corollary 1 (Overall Stability): The GAT-MATD3 framework constitutes a stable policy improvement algorithm. The iterative process of policy evaluation (Theorem 1) and policy improvement (Theorem 2) is guaranteed to converge, with the final performance bounded by the function approximation error and algorithmic bias.

Remark 3. We acknowledge that the assumptions of Lipschitz continuity and environmental quasi-stationarity used in Theorem 1 are strong abstractions compared to the highly dynamic V2X reality. In practice, these mathematical conditions are approximated via the architectural design. Specifically, the GAT module mitigates non-stationarity by transforming time-varying topological inputs into stable latent embeddings, thereby smoothing the effective state space. Similarly, the

Lipschitz property is approximated by the implicit Lipschitz-like regularity enforced by the neural network's architecture and bounded activation functions. Consequently, the convergence result guarantees the structural stability of the learning process under reasonable approximation, rather than claiming exact modeling fidelity in highly dynamic V2X channels.

To the best of our knowledge, this is the first work to establish formal convergence guarantees for an actor-critic method enhanced with graph attention in the context of multi-agent V2X resource allocation. These theoretical insights, combined with our extensive experimental validations in Sec. V, confirm that the proposed GAT-MATD3 is both practically effective and theoretically sound.

Algorithm 1 The GAT-MATD3 Algorithm

Input: Policy delay d_{policy} , soft update rate τ , target policy noise σ' , noise clip c , batch size B_s , learning rates α_A, α_C .

Output: Optimized actor network μ_ϕ .

Initialize shared actor network μ_ϕ and target network $\mu_{\phi'}$. Initialize critic networks $Q_{\theta_{i,k}}$ and target networks $Q_{\theta'_{i,k}}$ for each agent $i \in \mathcal{N}$ and $k \in \{1, 2\}$. Initialize replay buffer $\mathcal{D}$.

for each episode do

Observe initial joint observation $\mathbf{o}_1$ and construct graph $\mathcal{G}_1$. **for** $t = 1$ **to** $T_{\max}$ **do**

Select joint action with exploration noise: $\mathbf{a}_t \leftarrow \mu_\phi(\mathbf{o}_t, \mathcal{G}_t) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma)$.

Execute $\mathbf{a}_t$, receive rewards $\mathbf{r}_t$, done signal d_t , and next observation $\mathbf{o}_{t+1}$. Construct next graph $\mathcal{G}_{t+1}$.

Store transition $(\mathbf{o}_t, \mathbf{a}_t, \mathbf{r}_t, \mathbf{o}_{t+1}, d_t, \mathcal{G}_t, \mathcal{G}_{t+1})$ in $\mathcal{D}$.

if $|\mathcal{D}| > B_s$ **then**

Sample a mini-batch of B_s transitions from $\mathcal{D}$.

Compute target actions with smoothed noise:

$\bar{\mathbf{a}}' \leftarrow \mu_{\phi'}(\mathbf{o}', \mathcal{G}') + \text{clip}(\mathcal{N}(0, \sigma'), -c, c)$. **for each agent** $i = 1, \dots, N$ **do**

Compute the TD target y_i according to Eq. (20). Update critic parameters $\theta_{i,k}$ by one step of gradient descent on the loss from Eq. (21) for $k \in \{1, 2\}$.

if $t \pmod{d_{\text{policy}}} = 0$ **then**

for each agent $i = 1, \dots, N$ **do**

Compute current joint action from actor: $\mathbf{a} \leftarrow \mu_\phi(\mathbf{o}, \mathcal{G})$. Create a temporary action batch $\mathbf{a}_{\text{critic}}$ where actions a_j for $j \neq i$ are detached from the computation graph. Define actor loss for agent i : $J_i(\phi) \leftarrow -Q_{\theta_{i,1}}(\mathbf{o}, \mathbf{a}_{\text{critic}})$. Update shared actor parameters: $\phi \leftarrow \phi + \alpha_A \nabla_\phi J_i(\phi)$.

Perform soft updates on all target networks using Eqs. (24)-(25).

are implemented based on Python 3.11 and the PyTorch library. We construct a dynamic RIS-assisted V2X communication environment. The scenario is set in a 600m x 800m urban intersection, where the three-dimensional coordinates of the Base Station (BS) and the RIS are (500m, 800m, 25m) and (250m, 400m, 15m), respectively. The scenario involves $N=16$ mobile vehicles, forming 16 V2I links with the BS. Concurrently, each vehicle establishes a V2V link with its nearest neighbor. All V2I and V2V communications are composed of both direct and RIS-assisted reflected paths.

TABLE II: SIMULATION PARAMETERS

Parameter	Value
Carrier Frequency, f_c	2.0 GHz
Total System Bandwidth	1 MHz
Number of RBs, K	16
RB Bandwidth, W	62.5 kHz
Noise Power, σ^2	-114 dBm
BS Position	(500, 800) m
RIS Position	(250, 400) m
BS Height, H_{BS}	25 m
RIS Height, H_{RIS}	15 m
RIS Elements, M	64
Number of Vehicles, N	16
Vehicle Height, H_{veh}	1.5 m
Vehicle Speed Range	[10, 15] m/s
Turning Probability	0.4
Path Loss Model	3GPP UMi-Street Canyon
Fast Fading Model	Rayleigh Fading
Shadowing Decorrelation Dist.	15m (V2I), 10m (V2V)
Shadowing Std. (LoS/NLoS)	
V2I Links	4.0 dB / 6.0 dB
V2V Links	3.0 dB / 4.0 dB
V2V Power Range	[5, 23] dBm
V2I Transmit Power, P_{V2I}	23 dBm
V2V Payload Size, D_V	8480 bits (1060 Bytes)
Max. AoI Threshold, $A_{\max}$	10 s
Simulation Step, Δt	0.1 s
Reward Weight, w_A, w_P	10, 8

TABLE III: REINFORCEMENT LEARNING HYPERPARAMETERS

Hyperparameter	Value
Learning Rate, α_A and α_C	1×10^{-4}
Discount Factor, γ	0.99
Replay Buffer Size	50,000
Batch Size, B_s	512
Exploration Noise, σ	0.1
Target Network Update Rate, τ	0.005
Policy Delay, d_{policy}	2
Target Policy Noise, σ'	0.2
Noise Clip, c	0.5
Graph Conn. Threshold, d_{th}	150 m
Node Feature Dimension, D_h	64
GAT Hidden Dimension	64
GAT Attention Heads (1st Layer)	3
GAT Layers	2

V. NUMERICAL RESULTS

In this section, we evaluate the performance of our proposed GAT-MATD3 framework through simulations. The simulations

The vehicular mobility follows the Manhattan grid model implemented in our experiment: upon approaching an intersection, each vehicle has a 40% probability of making a

turn. The initial vehicle speed is randomly chosen from the interval [10, 15] m/s. At the beginning of each episode, we set the initial payload for V2V links to 8×1080 bits, and the initial AoI for V2I links is set to 100 ms. Our channel model strictly adheres to the 3GPP TR 38.901 standard, incorporating path loss, spatially correlated shadow fading, and Rayleigh fast fading. The neural network architectures are implemented as MLPs. Specifically, the actor network's encoder module uses two hidden layers of size [128, 128], while its decoder employs two hidden layers of [256, 256]. Each centralized critic network also consists of two hidden layers of size [256, 256]. The remaining key simulation and reinforcement learning hyperparameters are detailed in Table II and Table III, respectively.

To evaluate the performance of our method, we compare it against the following algorithms:

- **MATD3**: The foundation of our algorithm, used to directly isolate and evaluate the benefits of incorporating a GNN module for inter-agent communication. Its hyperparameters are adapted from [22].
- **GraphSAGE-MATD3**: A variant using a GraphSAGE-based GNN instead of GATv2. This baseline is designed to specifically test the effectiveness of our chosen dynamic attention mechanism against a more standard neighborhood aggregation method.
- **Centralized SAC**: A fully centralized Soft Actor-Critic agent that processes the concatenated global state and outputs a joint action for all vehicles. Its configuration is primarily based on [14], where SAC demonstrated superior performance compared to PPO, DDPG, and TD3. To ensure a rigorous comparison, we explicitly optimized the SAC implementation: we employed automatic entropy tuning to dynamically optimize the exploration-exploitation balance without manual scheduling. Furthermore, to mitigate the impact of the high-dimensional action space, we utilized a deep network architecture (three hidden layers of 512 units) with a large batch size of 1024, ensuring that the baseline's performance was not limited by network capacity or insufficient tuning.
- **Random Scheme**: A non-learning baseline where agents select actions based on a predefined heuristic random policy, rather than uniformly at random. This serves to establish a baseline performance level for a simple, uncoordinated strategy.

A. Behavioral Analysis of GAT-MARL Framework

To validate the learned policy, Fig. 3 provides a microscopic analysis of a representative decision-making instant at $T = 5$. The main 3D view depicts the macroscopic network topology of all 16 vehicles, the BS, and the RIS. The protagonist, Vehicle 3, operates in a congested environment where its neighbors, Vehicle 4 and Vehicle 11, were in conflict on RB 12 in the preceding time step ($T = 4$).

The top-left 2D inset visualizes the attention coefficients from the protagonist's GAT module. It allocates attention relatively uniformly across its neighbors (0.201 to Veh 4, 0.201 to Veh 11, 0.196 to Veh 9, and 0.200 to Veh 14), indicating

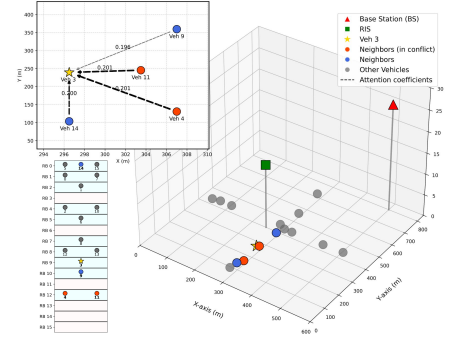


Fig. 3: Behavioral analysis of a representative decision-making instant.

that information is processed from all proximal vehicles in this contested state. These coefficients are normalized over all neighbors and the agent itself (self-attention was 0.2029). The bottom-left chart shows the resulting spectrum state at $T = 5$. RB-12 remains in a two-vehicle conflict, while RB-0 is occupied by three spatially distant vehicles, a valid spatial reuse under the K_{reuse} constraint. Based on this perception, Vehicle 3 selects the unoccupied RB-9. In summary, this case study illustrates the operational behavior of a trained agent. The selection of RB-9 results from an assessment of the agent's internal state, its perception of local risks such as the conflict on RB-12 versus the spatial reuse on RB-0, and the availability of global resources. This mapping from perception to action exemplifies the policy responsible for the macroscopic performance metrics reported.

B. Sensitivity and Robustness Analysis

To determine the optimal connection threshold d_{th} , we conducted a sensitivity analysis to balance interference visibility against computational complexity.

Physically, under the 3GPP UMi model (2 GHz), signal attenuation at 150 m is approximately 30 dB higher than at 20 m, approaching the thermal noise floor (-114 dBm). Thus, 150 m serves as a natural boundary to filter negligible interference. The 150 m threshold reflects 2 GHz UMi channel characteristics. Higher frequencies like the 5.9 GHz band exhibit faster attenuation, making this boundary more conservative. Conversely, highway environments or increased antenna heights extend the interference range. The stable performance across the 100–250 m range in Fig. 5 validates the framework's adaptability to such conditions. This physical hypothesis is empirically validated by the training results presented in Fig. 4. As shown, the value of $d_{th} = 150$ m yields a superior learning trajectory compared to other baselines, achieving the fastest convergence speed and the highest stable reward. Consequently, we adopt this value for all subsequent experiments.

To verify the final performance, we conducted a fixed-seed sensitivity analysis (Fig. 5). The results confirm that $d_{th} = 150$ m yields the lowest Remaining Demand Ratio (5.02%). Decreasing the threshold (100 m) causes information loss, while increasing it (200–250 m) introduces noise from distant nodes, both degrading reliability. Meanwhile, the stable

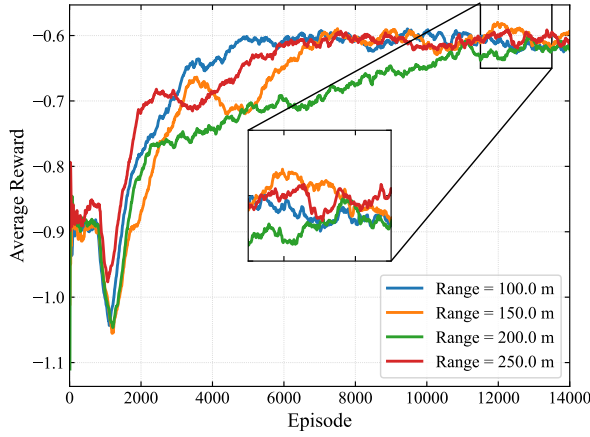


Fig. 4: Impact of connection threshold d_{th} on learning convergence.

AoI (≈ 3.64 ms) and consistent 100% PDR demonstrate the framework's robustness against parameter variations.

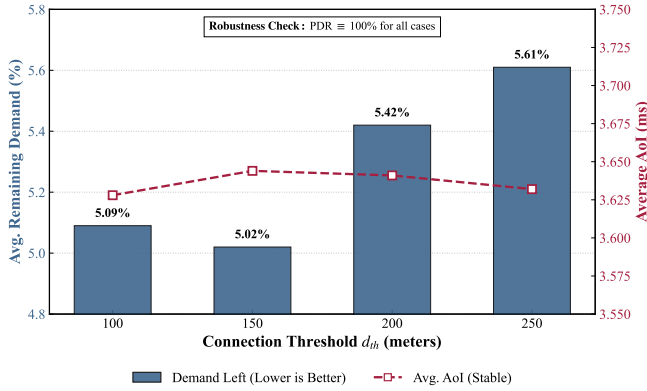


Fig. 5: Sensitivity analysis of system metrics.

To evaluate resilience against high-mobility impairments such as pilot contamination and Doppler aging, we introduced additive CSI noise. Under the AR-1 Jakes' model, temporal decorrelation leads to increased uncertainty in the observed CSI. The severe noise ($\sigma = 0.2$) in Table IV serves as a practical proxy for the level of channel uncertainty induced by these factors, which is consistent with a temporal correlation on the order of $\rho \approx 0.98$ under typical urban mobility (15 m/s). As shown, maintaining a 100% PDR and 3.74 ms AoI under such conditions confirms the robustness of the proposed agents against practical CSI degradations, without requiring an expansion of the MDP state space.

TABLE IV: Robustness against CSI Estimation Errors

Noise (σ)	PDR (%)	Avg. AoI (ms)	Demand (%)
0.0 (Perfect)	100.00	3.70	5.50
0.05	100.00	3.69	5.55
0.10	100.00	3.72	5.66
0.20 (Severe)	100.00	3.74	5.73

C. Performance Comparison and Analysis

Fig. 6 illustrates the convergence of the average reward, demonstrating the superior performance of our proposed GAT-

MATD3 framework in terms of both convergence speed and final policy quality. This improved performance stems from the adaptive cooperative perception capability of the GAT module. Specifically, the consistent performance gap between GAT-MATD3 and the non-cooperative MATD3 baseline highlights the importance of explicit information sharing for effective coordination. Furthermore, its outperformance of GraphSAGE-MATD3 validates the effectiveness of an adaptive attention mechanism over isotropic aggregation. The underperformance of the centralized SAC agent, despite its global observability, highlights a key limitation of centralized control. A single agent struggles to effectively navigate the exponentially large joint action space, leading to suboptimal policies. This confirms the practical advantage of our coordinated decentralized approach. The superior stability of GAT-MATD3 (as shown in Fig. 6) validates its resilience to device heterogeneity. While varying neighbor counts and channel conditions introduce variance, the GAT module's attention mechanism effectively standardizes these heterogeneous inputs to enhance learning accuracy and stability. Furthermore, the decentralized execution inherently maintains communication efficiency regardless of device density, preventing the performance collapse observed in the baseline.

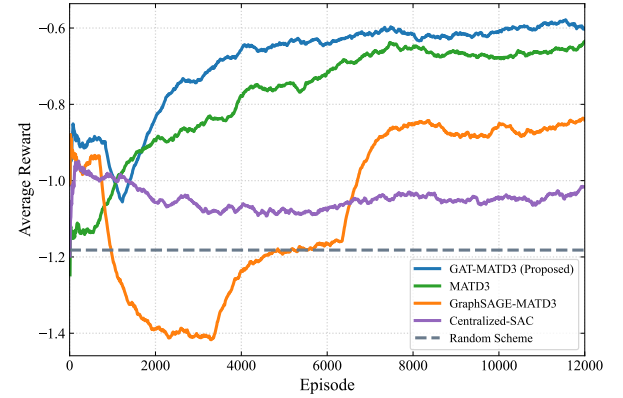


Fig. 6: Comparison of average reward convergence for different algorithms.

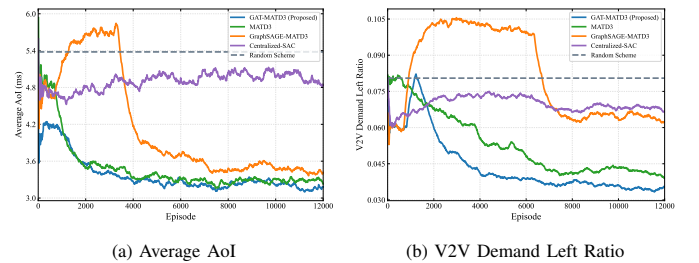


Fig. 7: Performance comparison on the primary optimization objectives.

To further deconstruct the performance of the algorithms, we separately examine the two primary optimization objectives: the average AoI for V2I links and the reliability of V2V links, which is measured by the V2V Demand Left Ratio.

Fig. 7 presents the convergence of these two core metrics, further validating the superiority of our framework. The results reveal that GAT-MATD3 learns a highly efficient and balanced resource allocation policy, concurrently achieving the lowest

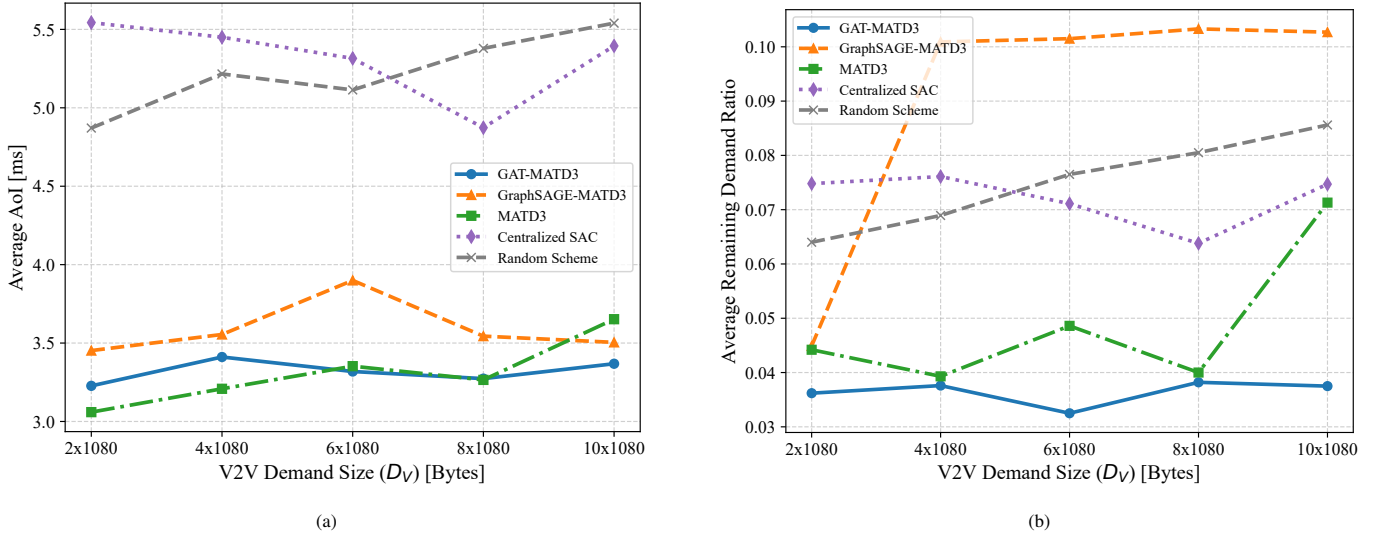


Fig. 8: Robustness analysis against varying V2V payload size (D_V). (a) Average AoI, (b) Average Remaining Demand Ratio.

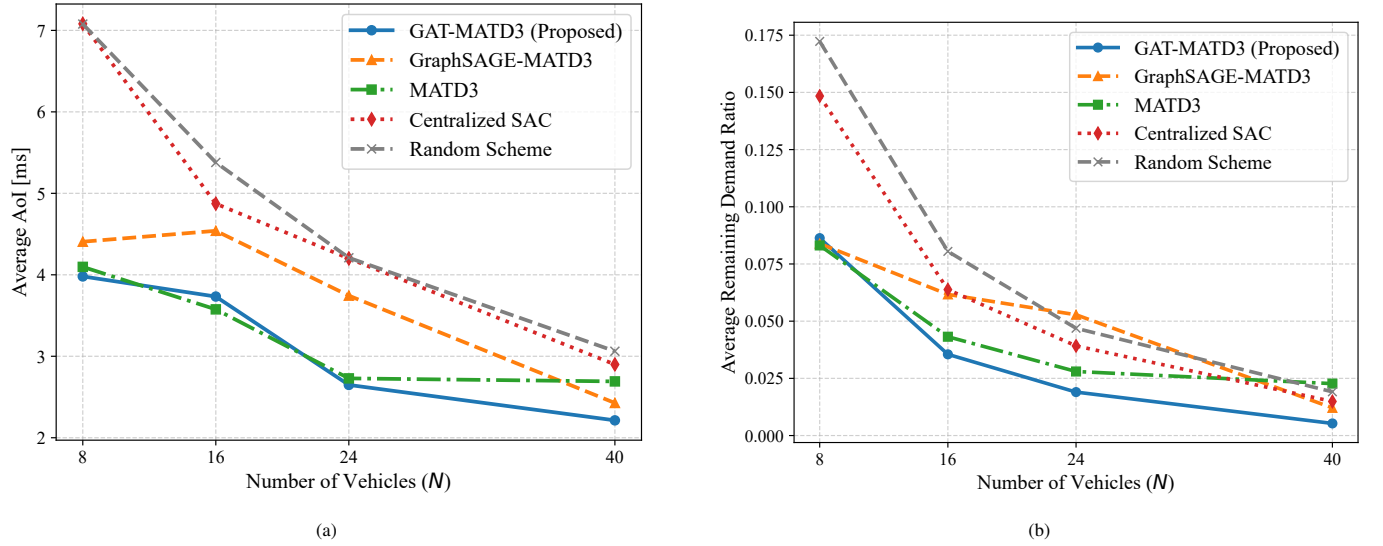


Fig. 9: Performance scalability analysis against varying number of vehicles (N). (a) Average AoI, (b) Average Remaining Demand Ratio.

values on both metrics and effectively navigating the Pareto front. As shown in Fig. 7(a), it minimizes the average AoI, ensuring optimal V2I timeliness, while simultaneously attaining the lowest V2V Demand Left Ratio for maximum reliability in Fig. 7(b). In contrast, while MATD3's performance follows a similar trend, a persistent gap remains on both metrics, highlighting the limitations of non-cooperative strategies. The other baselines struggle significantly, with GraphSAGE-MATD3 exhibiting unstable learning and Centralized-SAC becoming trapped in a mediocre equilibrium. This comprehensive analysis confirms that the superiority of GAT-MATD3 stems not from over-optimizing a single objective, but from its powerful multi-objective coordination capability, enabled by the accurate network state modeling of the GAT module.

D. Robustness and Scalability Analysis

In this section, we evaluate the robustness and scalability of our proposed framework by varying the V2V traffic load and

the number of vehicles in the network.

First, to assess the robustness against varying traffic demands, we increase the V2V payload size, D_V , fivefold from 2×1080 Bytes to 10×1080 Bytes, while keeping the number of vehicles fixed at $N = 16$. As shown in Fig. 8, the analysis clearly demonstrates the superior robustness of our proposed GAT-MATD3 algorithm. As the V2V traffic load intensifies, GAT-MATD3 maintains a consistently low and stable average AoI (Fig. 8a) and remaining demand ratio (Fig. 8b), indicating its ability to effectively adapt its resource allocation strategy without significant performance degradation. In contrast, all baseline algorithms exhibit deteriorating performance under heavier loads. MATD3, the second-best performer, shows a noticeable degradation, while GraphSAGE-MATD3's performance proves highly volatile, indicating its inability to manage the heightened resource contention. This analysis validates that the adaptive cooperative perception mechanism in GAT-MATD3 is crucial for maintaining stable network performance

in dynamic environments.

Next, to evaluate scalability, we vary the number of vehicles, N , across four representative scales: small ($N = 8$), medium ($N = 16$), large ($N = 24$), and extra-large ($N = 40$). The evaluated policies in Fig. 9 were explicitly trained from scratch at each respective scale, and the reported results are based on deterministic evaluations of the best-performing models saved during training. While larger state-action spaces at $N = 40$ introduce higher training variance for standard baselines, GAT-MATD3 maintains stable and efficient convergence by leveraging local graph aggregations to manage the representational burden. The results in Fig. 9 compellingly demonstrate that the advantages of our adaptive cooperation mechanism become increasingly crucial as the network densifies. Regarding AoI (Fig. 9a), GAT-MATD3 consistently outperforms all baselines, and its performance advantage widens with scale. In dense scenarios, the observed AoI reduction reflects the adaptive resource allocation behavior learned by the agents, where delay-sensitive V2I transmissions are prioritized under increased interference. In the most challenging $N = 40$ scenario, it achieves an AoI of 2.36ms, representing a significant 14.8% and 9.2% improvement over MATD3 and GraphSAGE-MATD3, respectively. This superiority is magnified in the V2V reliability metric (Fig. 9b), which is highly sensitive to effective cooperation. While all learning-based methods perform similarly in sparse networks ($N = 8$), their strategic differences diverge sharply as the network size increases. The non-cooperative MATD3 hits a performance ceiling, with its improvement decelerating after $N = 24$. In stark contrast, GAT-MATD3's performance continues to improve due to its efficient cooperation, culminating in a remaining demand ratio of just 1.02% in the extreme $N = 40$ scenario. This represents a nearly 60% reduction in task failure rate compared to MATD3 and a 40% improvement over the second-best performer, Centralized SAC.

E. Impact Analysis of RIS Scale

Finally, we analyze the impact of the RIS scale by varying the number of reflecting elements, M , from 16 to 128, while keeping the number of vehicles fixed at $N = 16$. This analysis, presented in Fig. 10, investigates how different algorithms leverage the reconfigurable wireless environment.

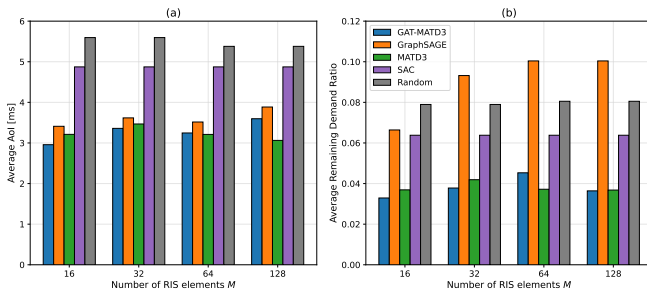


Fig. 10: Impact analysis of varying the number of RIS elements (M). (a) Average AoI. (b) Average Remaining Demand Ratio.

Fig. 10(b) illustrates the performance in terms of V2V reliability. Our proposed GAT-MATD3 consistently achieves

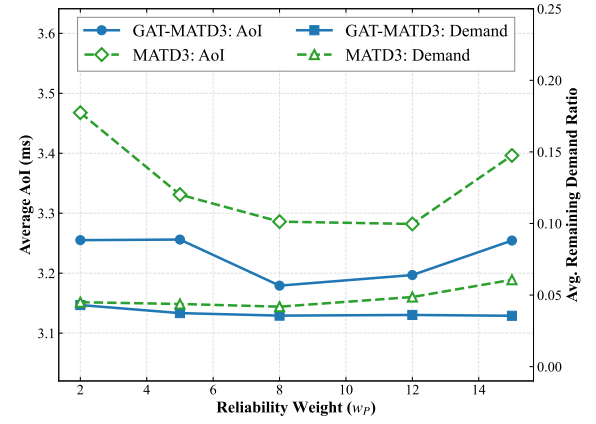


Fig. 11: Sensitivity analysis of the reliability weight w_P on system performance. The proposed GAT-MATD3 maintains robust performance across a wide range of weights compared to the baseline.

the lowest remaining demand ratio across all RIS scales, demonstrating its superior performance and stability. A particularly noteworthy finding is the significant performance degradation of GraphSAGE-MATD3 as M increases. This strongly suggests that its isotropic aggregation mechanism is ill-suited to exploit the complex channel improvements afforded by a larger RIS, potentially leading to counterproductive interference management. Regarding the AoI metric, Fig. 10(a) reveals a distinct trend. The AoI performance of both GAT-MATD3 and MATD3 remains highly stable and largely unaffected by the RIS scale, consistently maintaining a low value of approximately 3.2ms. This suggests that the agents have learned an intelligent resource prioritization strategy; once the V2I channel quality surpasses a sufficient threshold, even with a small RIS, strategic resources are preferentially allocated to the more challenging V2V reliability task. This behavior highlights the advanced multi-objective decision-making capability of our framework, showing that the framework can leverage physical layer improvements to enhance overall system performance.

F. Sensitivity Analysis of Reward Weights

Fig. 11 presents a sensitivity analysis on the reliability weight $w_P \in \{2, 5, 8, 12, 15\}$ with fixed $w_A = 10$. The proposed GAT-MATD3 exhibits superior robustness, maintaining consistently low Average AoI and Remaining Demand Ratio across the entire spectrum of weights. In contrast, the MATD3 baseline suffers significant performance degradation at lower reliability weights, particularly failing to effectively reduce the remaining demand backlog. These results demonstrate that the proposed framework maintains consistent superiority across a wide range of preference configurations, confirming that the observed performance gains are structural rather than sensitive to specific scalar weight choices.

G. Impact of RIS Control Strategy

To rigorously evaluate the robustness of the proposed GAT-MATD3 framework against various physical-layer configurations, we conducted a comparative analysis using three differ-

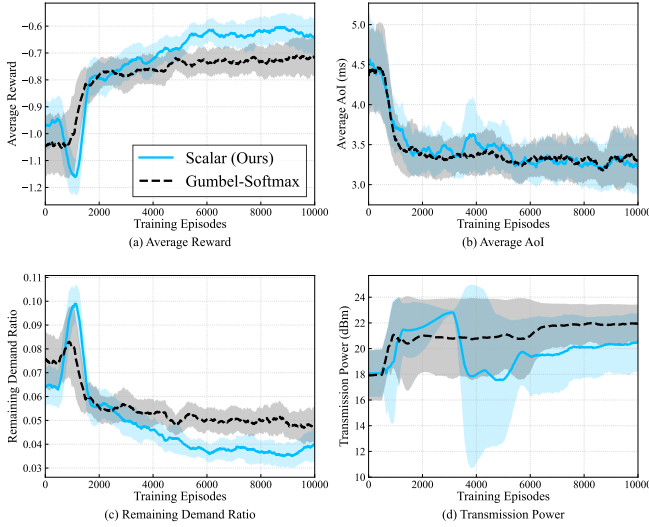


Fig. 12: Performance comparison of the proposed Scalar Mapping vs. Gumbel-Softmax Baseline, averaged over 3 random seeds with shaded standard deviation. The proposed method learns an aggressive strategy to clear data demand, whereas the baseline gets stuck in a conservative local optimum due to scalability limitations.

ent RIS control strategies: a fully Random RIS, a Codebook-based RIS, and our adopted Heuristic RIS.

TABLE V: RIS Control Strategy Impact Analysis (GAT-MATD3)

Performance Metric	Random RIS	Codebook RIS	Heuristic RIS (Proposed)
Average AoI (ms)	3.3566	3.0754	3.2007
Demand Left Ratio	3.84%	3.44%	3.46%
PDR (Success Rate)	100.00%	100.00%	100.00%

As Table V shows, the optimized Codebook RIS achieves the lowest Average AoI (3.08 ms) and Demand Left Ratio (3.44%). Our low-complexity Heuristic scheme yields highly comparable performance (3.20 ms and 3.46%), whereas the unoptimized Random baseline performs the worst. Despite such physical-layer capacity variations, GAT-MATD3 consistently maintains a 100% Packet Delivery Rate (PDR) across all scenarios. These results suggest that our MARL agents can compensate for suboptimal RIS configurations through adaptive cooperative resource allocation, highlighting the robustness of the framework against physical-layer uncertainties.

H. Comparison with Differentiable Policy

Fig. 12 provides an empirical comparison between the proposed scalar mapping and the Gumbel-Softmax baseline. As observed, the Gumbel-Softmax baseline exhibits premature convergence to a conservative local optimum, particularly in the considered multi-agent setting with a high-dimensional joint action space. In contrast, our method adopts a pragmatic relaxation that prioritizes practical scalability and training stability over strict theoretical gradient fidelity. This design choice enables more effective exploration and allows the agent to learn an aggressive policy that better satisfies QoS requirements, resulting in higher asymptotic rewards and a significantly reduced data backlog.

VI. CONCLUSION

This paper has presented GAT-MARL, a novel multi-agent reinforcement learning framework designed to resolve the fundamental trade-off between performance and scalability in resource allocation for RIS-assisted V2X networks. The core of our contribution is a bespoke GAT integrated within a powerful MATD3-based CTDE architecture. This design directly addresses the critical limitation of partial observability in decentralized systems by enabling agents to learn an adaptive cooperative perception policy, facilitating sophisticated coordination that transcends the capabilities of conventional MARL approaches. Our extensive simulations and analyses demonstrate the effectiveness and performance advantages of the proposed GAT-MARL framework. In direct comparison with strong baselines, including non-cooperative MATD3 and a centralized SAC agent, GAT-MARL demonstrated not only significantly accelerated convergence but also a higher final policy. This performance is manifested in its profound multi-objective optimization capability, concurrently minimizing the average AoI for V2I links and maximizing V2V reliability. In the most challenging large-scale scenario ($N = 40$), our method achieved a 14.8% improvement in AoI over the next-best MARL baseline and reduced the V2V task failure rate by nearly 60% compared to a non-cooperative approach, highlighting the effectiveness of its attention-based coordination mechanism. The practical viability of our framework was further substantiated through rigorous scalability and robustness analyses. Crucially, the performance advantage of GAT-MARL became more pronounced in denser networks and under heavier traffic loads, highlighting its suitability for the complex, large-scale V2X systems of the future. Moreover, our findings indicate that GAT-MARL intelligently exploits the reconfigurable wireless environment, strategically leveraging RIS-induced channel gains to enhance V2V reliability without compromising V2I timeliness. Finally, regarding the optimization of the RIS itself, while this work prioritized a decoupled design to maintain the focus on decentralized vehicular cooperation and ensure low-latency execution, we identify the joint optimization of MARL-based resource allocation and dynamic RIS control as a promising direction. To address this challenge, we plan to reserve the design of heterogeneous agent architectures for our future work, aiming to coordinate fast-timescale vehicular agents with slow-timescale RIS controllers.

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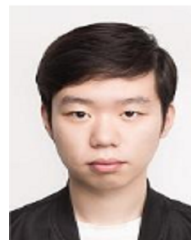
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